

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: Artificial Intelligence
CSA17**

COURSE CODE:

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.*
(BTL 5)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Reverse Engineering

1. System Decomposition & Analysis

Analyze an existing AI-based system and **reverse engineer its architecture**, identifying key components, data flow, and underlying algorithms.

2. Architecture Reconstruction

Given a black-box software system, **reconstruct its system architecture** and explain the interaction between modules using appropriate diagrams.

3. Algorithm Identification

Reverse engineer a machine learning application to **identify the algorithms, data processing steps, and decision-making logic** used in the system.

4. Data Flow & Process Mapping

Examine an existing distributed AI application and **map its data flow, processing pipeline, and communication between components**.

5. Improvement & Redesign

Perform reverse engineering on an intelligent system and **propose enhancements or redesign strategies** to improve scalability, efficiency, and performance.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. System Decomposition & Analysis

Example System: AI-Based Fraud Detection System

Objective

The purpose of the system is to analyze financial transactions and identify suspicious or fraudulent activities.

System Decomposition

The system can be divided into the following components:

1. Data Collection Module

- Collects transaction data.
- Examples: amount, time, location, account ID, and device information.

2. Data Preprocessing Module

- Removes incomplete or duplicate records.
- Converts raw data into a suitable format.

3. Feature Extraction Module

- Extracts important features such as:
 - Transaction frequency
 - Transaction amount
 - Location changes
 - Unusual user behavior

4. Machine Learning Module

- Uses an anomaly detection or classification algorithm.
- Calculates the probability of fraud.

5. Decision-Making Module

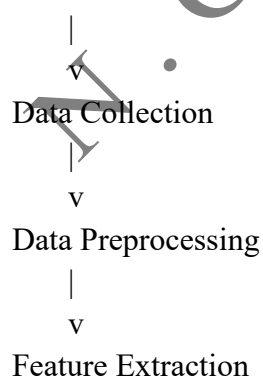
- Classifies transactions as:
 - Normal
 - Suspicious
 - Fraudulent

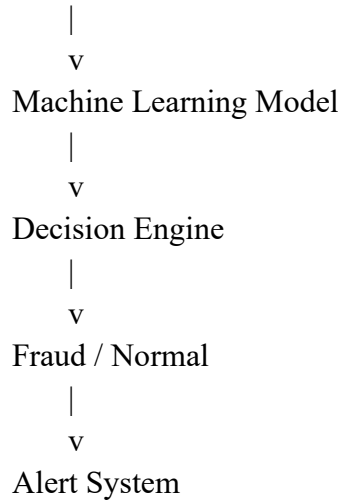
6. Alert Module

- Sends alerts when suspicious activity is detected.

Architecture Diagram

Transaction Data





Data Flow

Input Data



Cleaning



Feature Extraction



Model Prediction



Decision



Alert / Result

Underlying Algorithms

Possible algorithms include:

- Decision Tree
- Random Forest
- Logistic Regression
- Isolation Forest
- K-Means Clustering

Conclusion

By reverse engineering the fraud detection system, we can identify its major components, understand the data flow, and determine the algorithms used for detecting suspicious transactions.

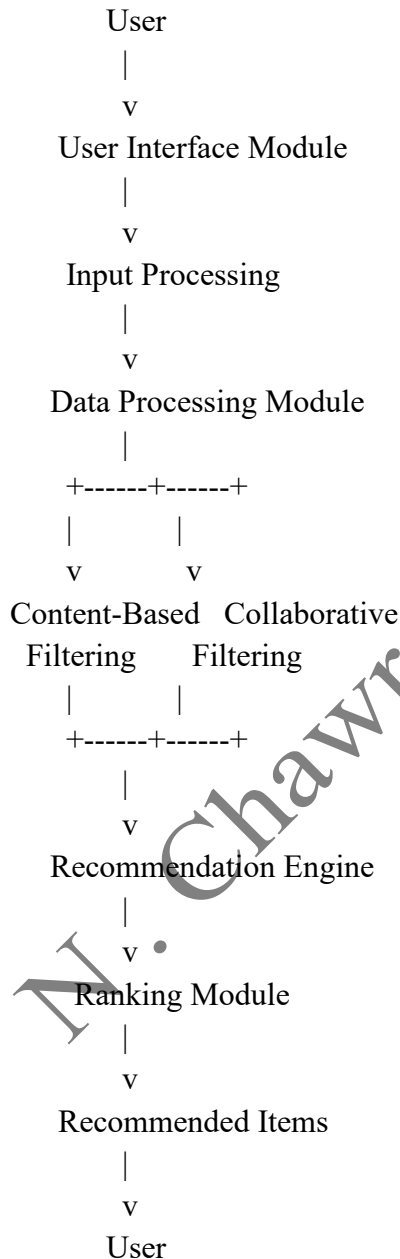
2. Architecture Reconstruction

Example: Black-Box AI-Based Recommendation System

A black-box system does not provide information about its internal architecture. We can reconstruct the architecture by observing:

- Inputs
- Outputs
- System behavior
- Response time
- User interactions

Reconstructed Architecture



Module Interaction

User Interface Module

Receives user actions such as:

- Search
- Click
- Rating
- Purchase

Data Processing Module

Processes user and product information.

Recommendation Modules

Generate recommendations using:

- User preferences
- Similar products
- Similar users

Ranking Module

Ranks recommendations according to relevance.

Reverse Engineering Method

Observe Input



Observe Output



Identify System Behavior



Divide System into Modules



Determine Module Interaction



Reconstruct Architecture

Conclusion

By analyzing the inputs and outputs of a black-box recommendation system, its internal modules and interactions can be reconstructed.

3. Algorithm Identification

Example: Machine Learning-Based Spam Detection System

The goal is to reverse engineer the application and identify the algorithm and decision-making logic.

Step 1: Identify Input

The system receives:

- Email subject
- Email content
- Sender information
- Number of links
- Keywords

Step 2: Identify Data Processing

Raw Email



Text Cleaning



Remove Stop Words



Tokenization



Feature Extraction



Numerical Feature Vector

Step 3: Identify Possible Algorithms

By analyzing the output and behavior, possible algorithms include:

- Naive Bayes
- Logistic Regression
- Support Vector Machine
- Decision Tree

A common approach is **Naive Bayes**, where the probability of spam is calculated using words and features.

Decision-Making Logic

Email Features



Machine Learning Model



Calculate Spam Probability



Probability > Threshold? No



Yes



Conclusion

Reverse engineering helps identify the data preprocessing steps, feature extraction methods, machine learning algorithm, and decision logic used by the application.

4. Data Flow & Process Mapping

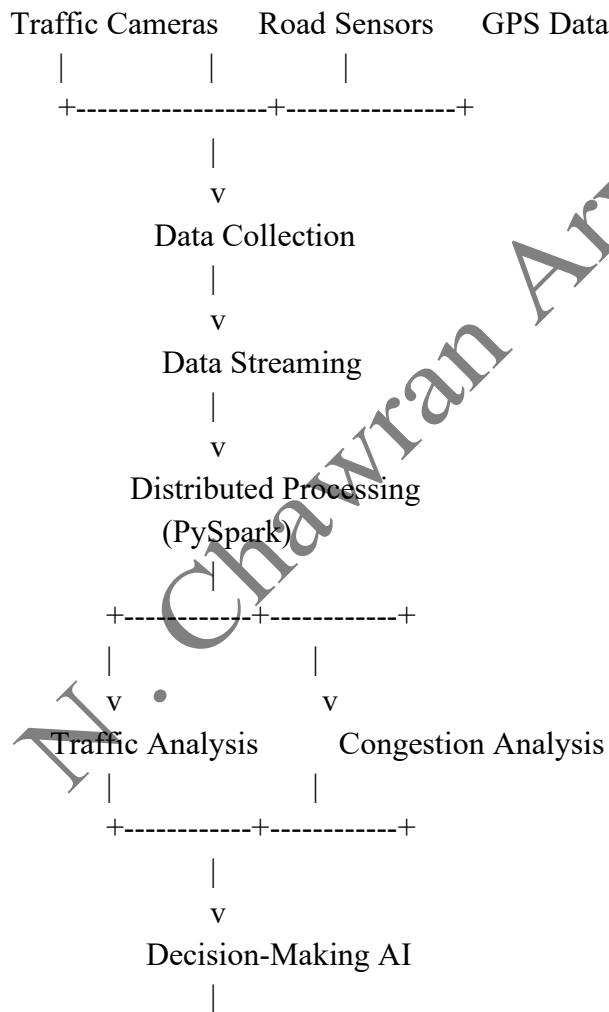
Example: Distributed AI-Based Smart Traffic System

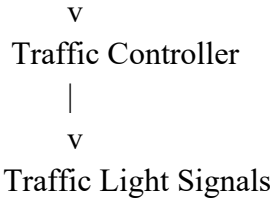
The system analyzes real-time traffic information from multiple sources.

Data Sources

- Traffic cameras
- Road sensors
- GPS data
- Vehicle information

Data Flow Diagram





Processing Pipeline

Step 1: Data Collection

Traffic information is continuously collected.

Step 2: Data Streaming

The collected information is sent to the distributed processing system.

Step 3: Data Processing

PySpark processes large amounts of data in parallel.

Example RDD operations:

map() → Transform traffic records

filter() → Remove invalid records

reduceByKey() → Calculate traffic density

groupByKey() → Group data by road

Step 4: AI Analysis

The AI system identifies congestion and abnormal traffic conditions.

Step 5: Decision

The system decides the appropriate traffic signal duration.

Step 6: Action

The traffic controller changes the traffic signals.

Conclusion

The data flow mapping shows how information travels from sensors to distributed processing, AI analysis, decision-making, and finally to the traffic control system.

5. Improvement & Redesign

Example: Improving an Intelligent Fraud Detection System

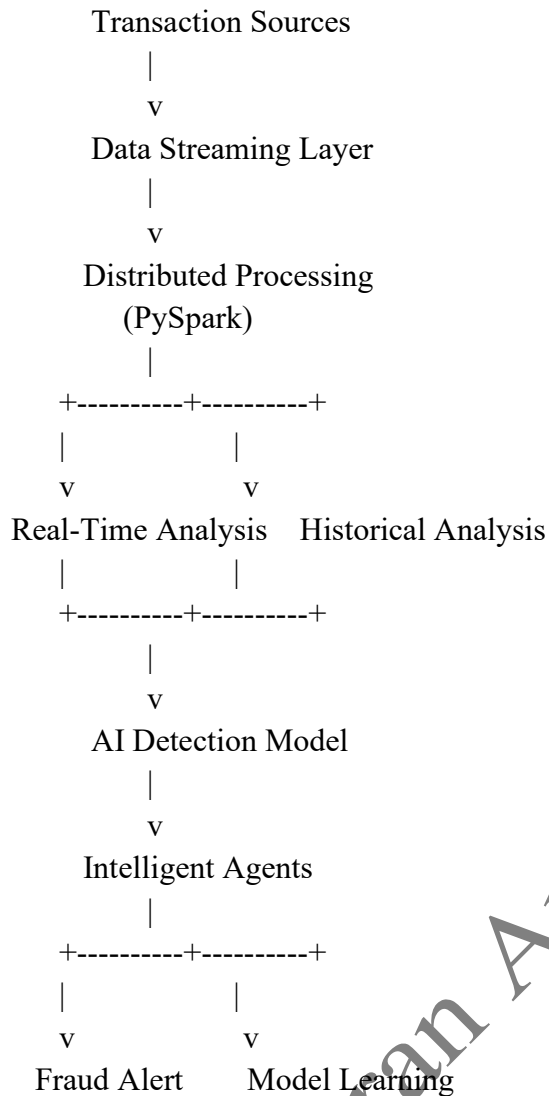
After reverse engineering the existing system, several improvements can be proposed.

Existing System Problems

Possible limitations include:

- Slow processing of large transaction volumes
- Limited scalability
- High false-positive rate
- Delayed fraud alerts
- Single centralized processing system

Proposed Redesigned Architecture



Proposed Improvements

1. Distributed Processing

Use PySpark to distribute transaction data across multiple nodes.

Benefits:

- Faster processing
- Parallel computation
- Better scalability

2. Real-Time Data Processing

Introduce a streaming layer to detect fraud immediately instead of processing data only in batches.

3. Intelligent Agents

Use multiple agents:

- **Monitoring Agent** – monitors transactions.
- **Detection Agent** – identifies anomalies.

- **Alert Agent** – sends warnings.
- **Learning Agent** – improves the model using feedback.

4. Model Improvement

Use feedback from confirmed fraud cases to improve detection accuracy and reduce false positives.

5. Scalable Architecture

More Data



Add More Worker Nodes



More Parallel Processing



Better Performance

Conclusion

The redesigned system improves scalability, efficiency, and performance by using distributed processing, real-time analysis, intelligent agents, and continuous model improvement.